

TM Forum Component

Service Performance Management

TMFC037

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Notice

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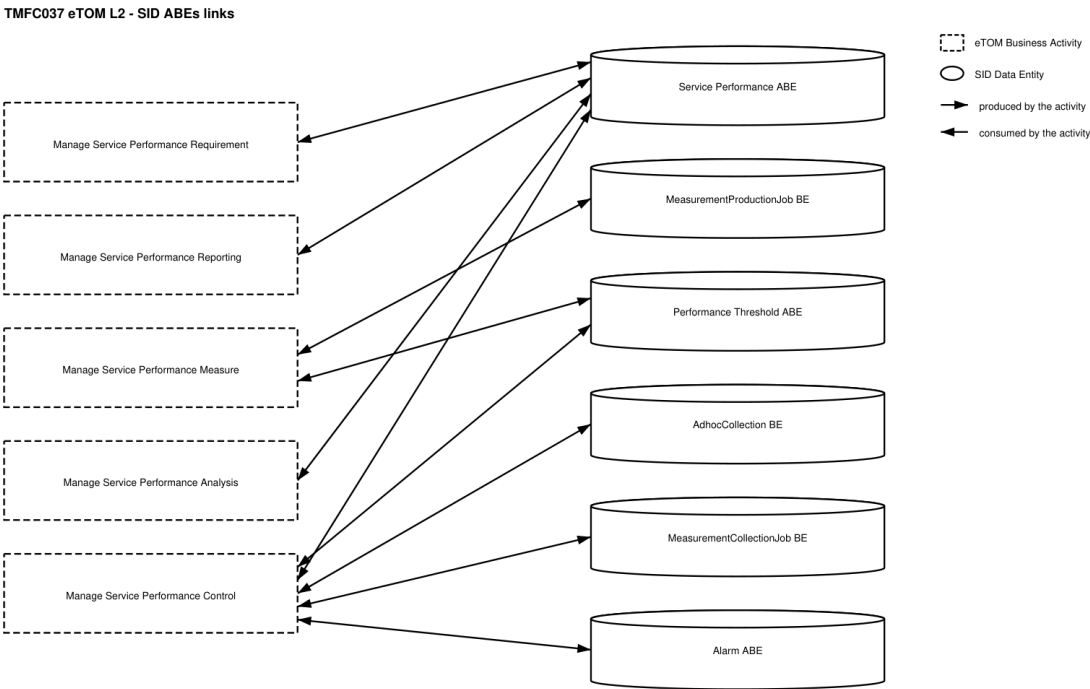
TMFC037 – Service Performance Management

1. Overview

Component Name	ID	Description	ODA Function Block
Service Performance Management	TMFC037	Service Performance Management component monitors, analyzes, and reports on end-to-end service performance. This can include a real-time, end-to-end view to ensure that each service is functioning correctly as well as a historical view. These functions build on the Resource Performance data and active end-to-end service performance test data to provide a view of a service. The component provides a key input to determine Quality of Service.	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
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1.4.7	L2	Service Performance Management	Service Performance Management business process directs and controls activities that define Service Performance Objectives (e.g. Service Availability, Service Quality, Process efficiency, Service Reliability, etc.), sets performance goals & targets, track performance trends, monitor performance, analyze performance, control performance (optimize, troubleshoot), report or communicate performance, and manage consequences of service performance. Service Performance Management business process supports the Enterprise Performance Management goals. These goals include quality, efficiency, reliability, availability, monetary (cost, profitability, etc.) mandates along with any productivity requirement of service delivery and service engagement by the organization. It includes identifying, establishing the applying the methodologies that define and manage service performance criteria to meet business objectives. / This process was renamed in 23.5 it was named Service Quality Management.
1.4.7.6	L3	Manage Service Performance Requirement	Manage Service Performance Requirement business activity controls activities and underlying tasks that define the performance requirements for a service. It sets the standards and expectations for service performance. / Manage Service Performance Requirement's goal is to ensure that the service meets or exceeds predefined performance standards. It involves setting service performance goals and making changes to these goals as necessary. It also includes defining the information required to measure service performance.

1.4.7.7	L3	Manage Service Performance Plan	Manage Service Performance Plan business activity controls the creating, implementing, improving, and terminating of plan(s) for service performance. / Manage Service Performance Plan's goal is to ensure that there is a clear and effective plan in place for managing service performance. This includes defining the plan, implementing it, making improvements to the plan over time, and terminating the plan when it is no longer needed.
1.4.7.8	L3	Manage Service Performance Measure	Manage Service Performance Measure business activity controls the definition, change, and removal of measures for service performance.
1.4.7.9	L3	Manage Service Performance Analysis	Manage Service Performance Analysis business activity controls activities and tasks supporting assessment, investigating, auditing, and reporting on service performance.
1.4.7.10	L3	Manage Service Performance Control	Manage Service Performance Control business activity is responsible for monitoring, evaluating, and improving service performance. / Manage Service Performance Control's goal is to maintain control over service performance. This includes monitoring performance on an ongoing basis, evaluating whether performance goals are being met, and providing support for making improvements as necessary.

1.4.7.11	L3	Manage Service Performance Reporting	Manage Service Performance Reporting business activity is responsible for creating, changing, and publishing reports on service performance. / Manage Service Performance Reporting's goal is to communicate information about service performance to relevant stakeholders. This includes creating reports that accurately reflect current performance levels, making changes to these reports as necessary, and publishing these reports in a timely manner.
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2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition
Service Performance ABE	The Service Performance ABE collects information used to correlate, consolidate, and validate various performance statistics and other operational characteristics of Know-Hows (a.k.a. CFS) and Technical Solutions (a.k.a. RFS). It provides a set of entities that enables performance monitoring and reporting. Each of these entities also enables network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Service degradation. Entities in this ABE also enable the traffic trend analysis.		<i>(no description available)</i>
Performance ABE	<i>(no description available)</i>	MeasurementProductionJob BE	<i>(no description available)</i>

Performance ABE	(no description available)	AdhocCollection BE	(no description available)
Performance ABE	(no description available)	MeasurementCollectionJob BE	(no description available)
Performance ABE	The Performance ABE is a pattern that represents a measure of the manner in which a Product and/or Service and/or Resource is functioning.	Performance Threshold ABE	(no description available)
Resource Trouble ABE	The Resource Trouble ABE collects information about problems found in allocated resource instances, regardless of whether the problem is physical or logical. Entities in this ABE capture detection, root cause, and resolution of these problems and maintain a history of the activities involved in diagnosing and solving the problem. This history includes tracking, reporting, assigning people to fix the problem, testing and verification, and overall administration of repair activities.	Alarm ABE	(no description available)

2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function L1	Aggregate Function L2	Function Description
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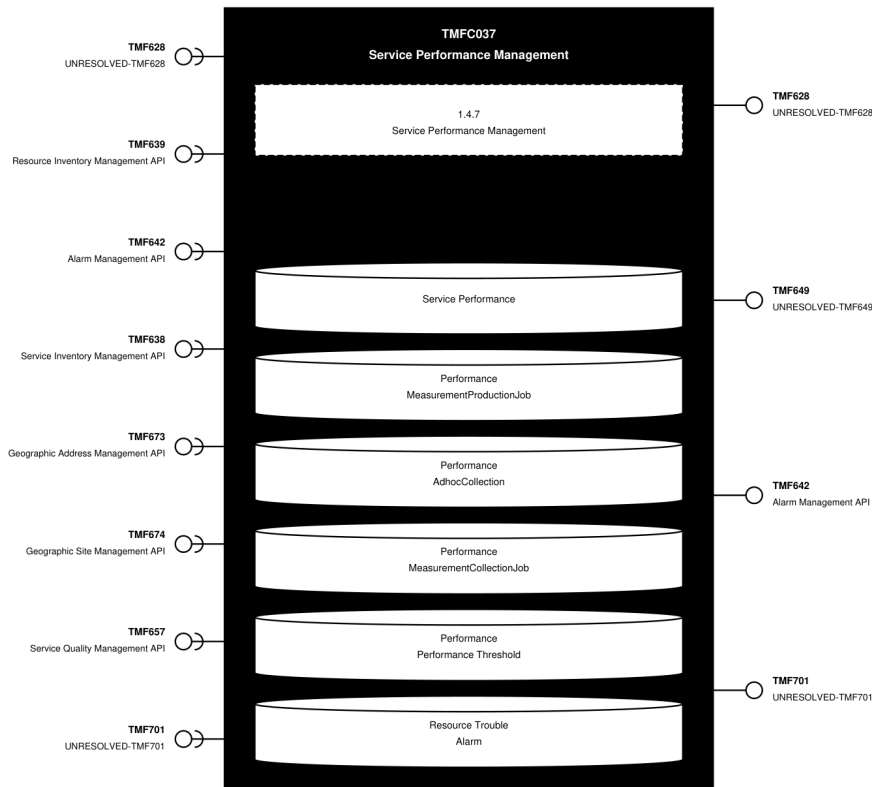
603	Service Performance Data Collection	Service Quality and Performance Management	Service Quality and Performance Inventory Management	Service Performance Data Collection collects Service Performance Data from the Resource Management functions related to a specific Service (or directly in the absence of Resource Performance Management function) and Collection of Service Performance data from the end-to-end tests done by the service test functions internally or from external Service Test applications. Including archiving.
604	Service Performance Event Correlation	Service Quality and Performance Management	Service Quality and Performance Analysis	Service Performance Event Correlation maps the performance data to service topology and identifies service related performance problems, event filtering included.
605	Service Performance Monitoring	Service Quality and Performance Management	Service Quality and Performance Analysis	Service Performance Monitoring provides monitoring of service performance data including notification and accumulation for e.g. Service Performance Dashboard.

606	Service Performance Reporting	<i>(no description available)</i>	Service Quality and Performance Inventory Management	Service Performance Reporting provides the necessary functionality required to generate reports about the performance of the service provider's services. These reports may be generated as part of the normal periodic operations ("scheduled") or may be as a result of a specific analysis request ("in-demand"). Report types include near real time, historical view, and trend analysis. Relevant performance reports are also provided to service/network planning to perform network updates.
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3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

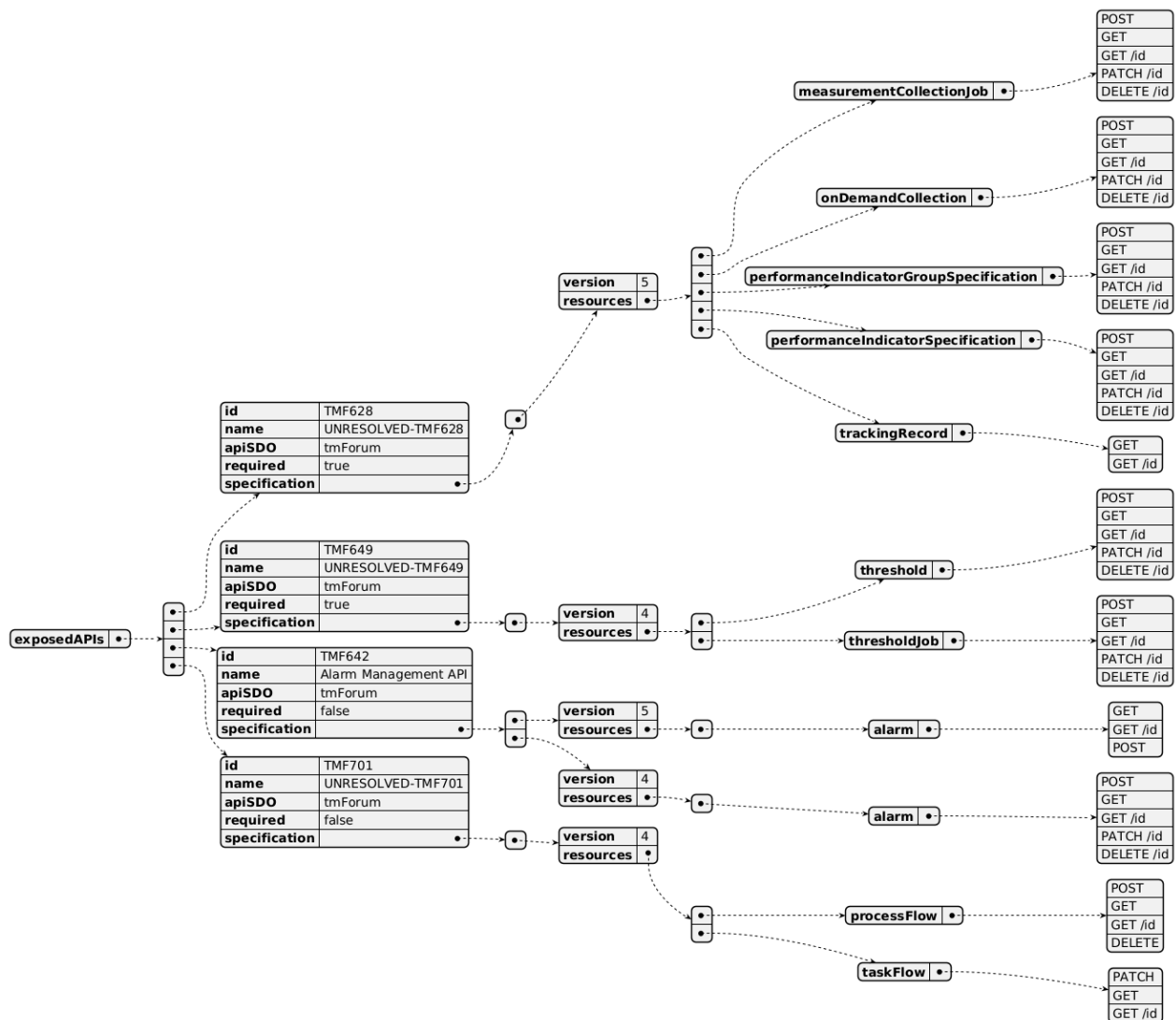


(SVG source: [TMFC037_API_Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the component-specification-documentation skill's scripts/render_api_context_svg.py.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF628	UNRESOLVED-TMF628	Mandatory	5	measurementCollectionJob	POST, GET, GET /id, PATCH /id, DELETE /id
TMF628	UNRESOLVED-TMF628	Mandatory	5	onDemandCollection	POST, GET, GET /id, PATCH /id, DELETE /id
TMF628	UNRESOLVED-TMF628	Mandatory	5	performanceIndicatorGroupSpecification	POST, GET, GET /id, PATCH /id, DELETE /id

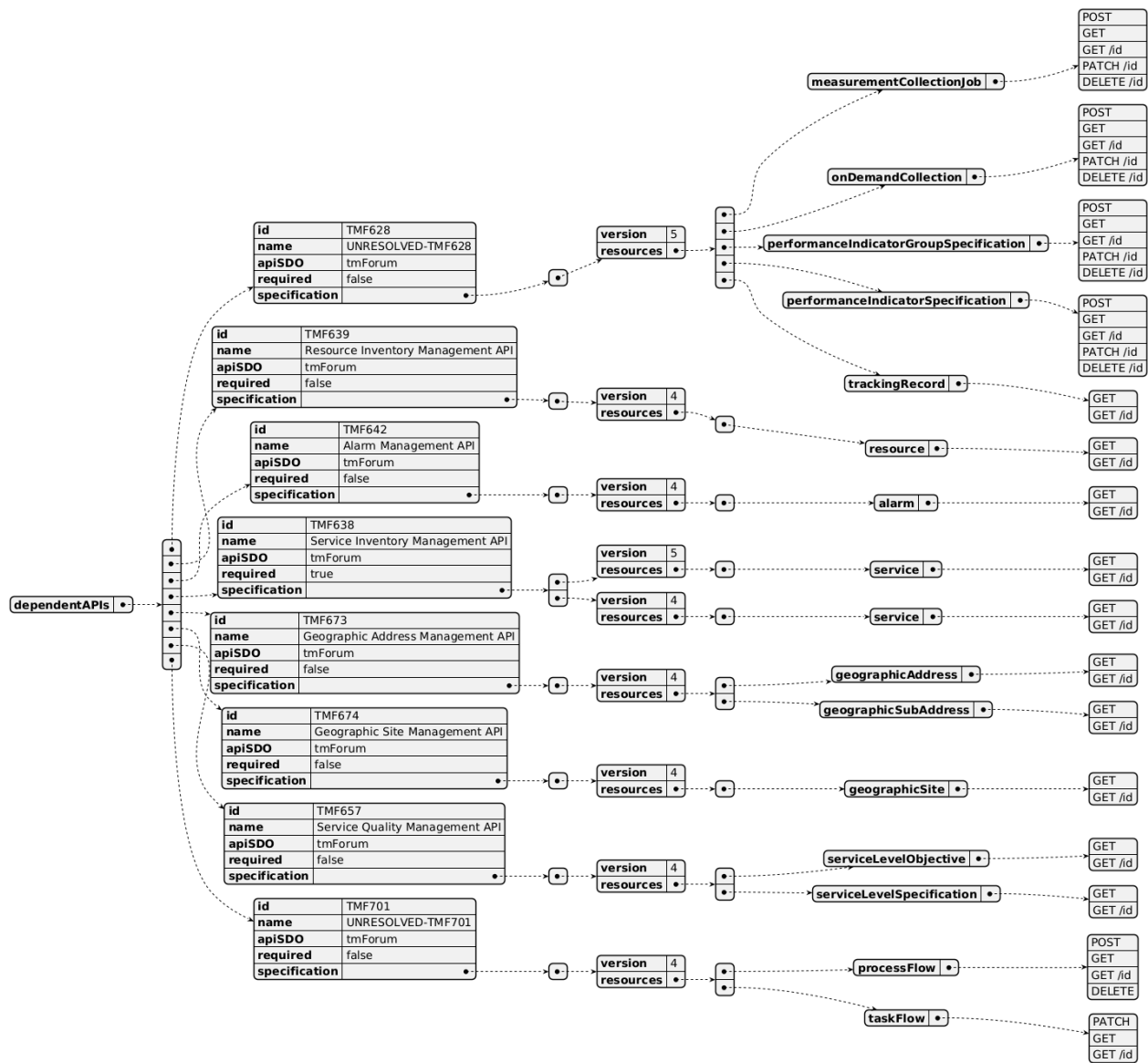
TMF628	UNRESOLVED-TMF628	Mandatory	5	performanceIndicatorSpecification	POST, GET, GET /id, PATCH /id, DELETE /id
TMF628	UNRESOLVED-TMF628	Mandatory	5	trackingRecord	GET, GET /id
TMF649	UNRESOLVED-TMF649	Mandatory	4	threshold	POST, GET, GET /id, PATCH /id, DELETE /id
TMF649	UNRESOLVED-TMF649	Mandatory	4	thresholdJob	POST, GET, GET /id, PATCH /id, DELETE /id
TMF642	Alarm Management API	Optional	5	alarm	GET, GET /id, POST
TMF642	Alarm Management API	Optional	4	alarm	POST, GET, GET /id, PATCH /id, DELETE /id
TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	PATCH, GET, GET /id



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF628	UNRESOLVED-TMF628	5	Optional	measurementCollectionJob	POST, GET, GET /id, PATCH /id, DELETE /id
TMF628	UNRESOLVED-TMF628	5	Optional	onDemandCollection	POST, GET, GET /id, PATCH /id, DELETE /id
TMF628	UNRESOLVED-TMF628	5	Optional	performanceIndicatorGroupSpecification	POST, GET, GET /id, PATCH /id, DELETE /id

TMF628	UNRESOLVED-TMF628	5	Optional	performanceIndicatorSpec	POST, GET, GET /id, PATCH /id, DELETE /id
TMF628	UNRESOLVED-TMF628	5	Optional	trackingRecord	GET, GET /id
TMF639	Resource Inventory Management API	4	Optional	resource	GET, GET /id
TMF642	Alarm Management API	4	Optional	alarm	GET, GET /id
TMF638	Service Inventory Management API	5	Mandatory	service	GET, GET /id
TMF638	Service Inventory Management API	4	Mandatory	service	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicAddress	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicSubAddress	GET, GET /id
TMF674	Geographic Site Management API	4	Optional	geographicSite	GET, GET /id
TMF657	Service Quality Management API	4	Optional	serviceLevelObjective	GET, GET /id
TMF657	Service Quality Management API	4	Optional	serviceLevelSpecification	GET, GET /id
TMF701	UNRESOLVED-TMF701	4	Optional	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	4	Optional	taskFlow	PATCH, GET, GET /id



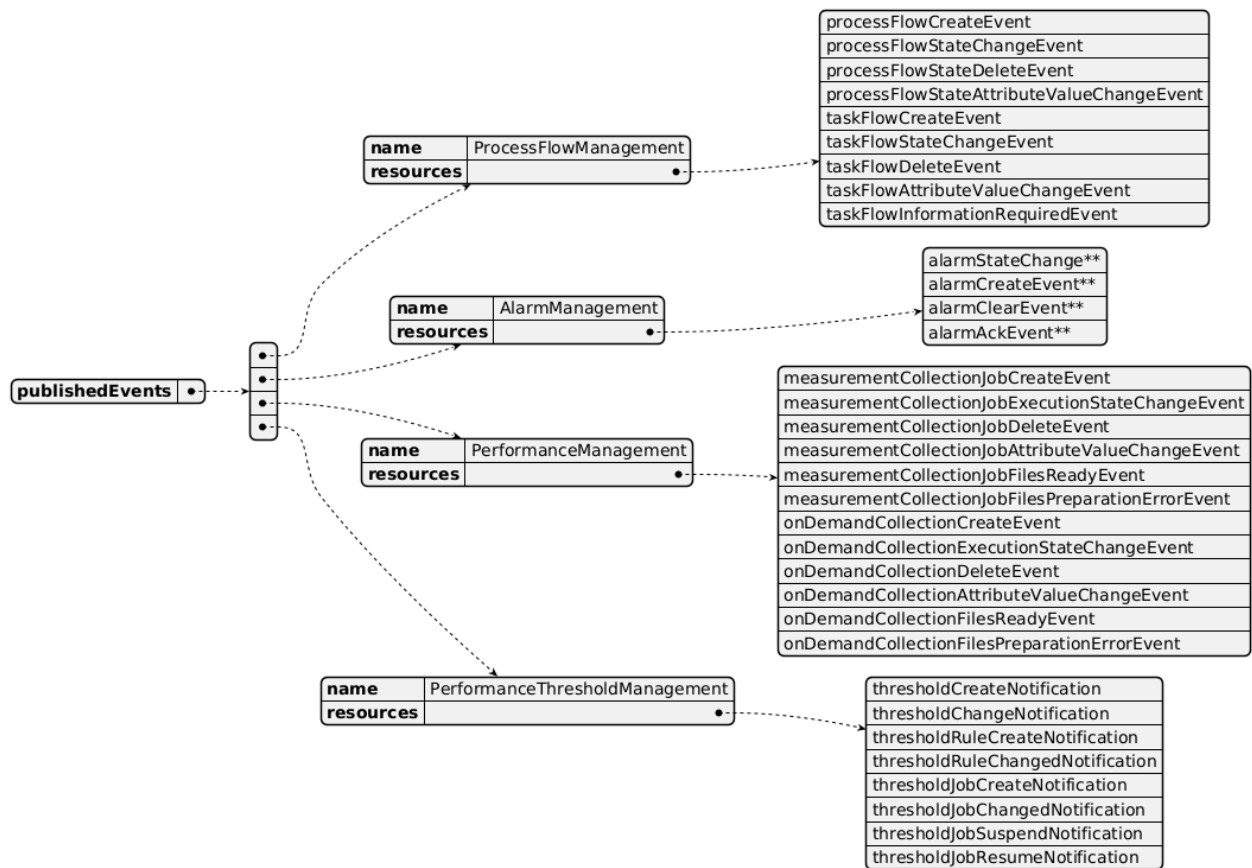
3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

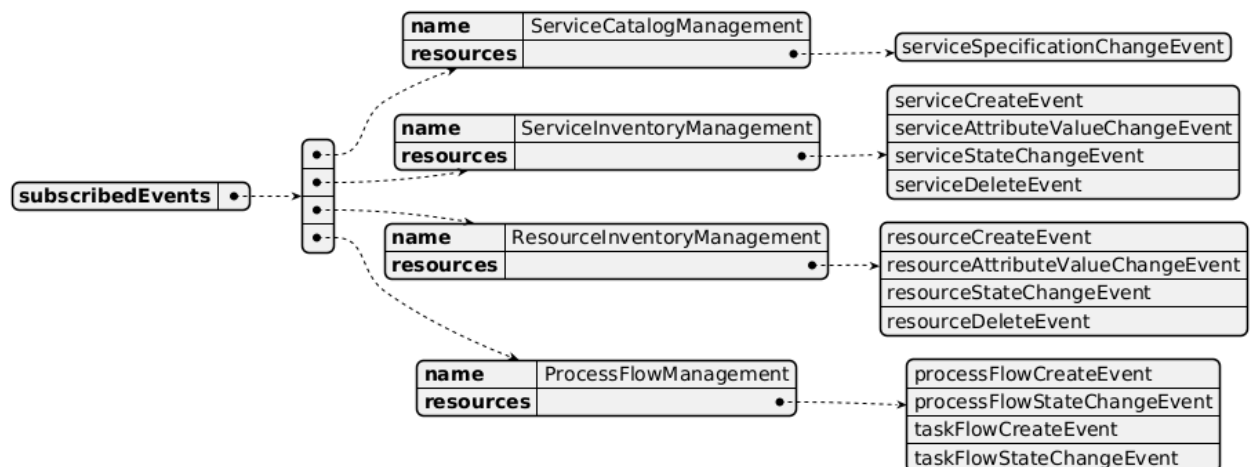
API ID	API Name	Event Resources
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TMF701	UNRESOLVED-TMF701	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent
TMF642	Alarm Management API	alarmStateChange alarmCreateEvent alarmClearEvent alarmAckEvent
TMF628	UNRESOLVED-TMF628	measurementCollectionJobCreateEvent measurementCollectionJobExecutionStateChangeEvent measurementCollectionJobDeleteEvent measurementCollectionJobAttributeValueChangeEvent measurementCollectionJobFilesReadyEvent measurementCollectionJobFilesPreparationErrorEvent onDemandCollectionCreateEvent onDemandCollectionExecutionStateChangeEvent onDemandCollectionDeleteEvent onDemandCollectionAttributeValueChangeEvent onDemandCollectionFilesReadyEvent onDemandCollectionFilesPreparationErrorEvent
TMF649	UNRESOLVED-TMF649	thresholdCreateNotification thresholdChangeNotification thresholdRuleCreateNotification thresholdRuleChangedNotification thresholdJobCreateNotification thresholdJobChangedNotification thresholdJobSuspendNotification thresholdJobResumeNotification



Subscribed Events

API ID	API Name	Event Resources
TMF633	Service Catalog Management API	serviceSpecificationChangeEvent
TMF638	Service Inventory Management API	serviceCreateEvent serviceAttributeValueChangeEvent serviceStateChangeEvent serviceDeleteEvent
TMF639	Resource Inventory Management API	resourceCreateEvent resourceAttributeValueChangeEvent resourceStateChangeEvent resourceDeleteEvent
TMF701	UNRESOLVED-TMF701	processFlowCreateEvent processFlowStateChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v25.0
SID	v25.0
Functional Framework	v25.0

5.2. Jira References

eTOM (*none currently listed*)

SID - ISA-693 Alarm ABE is currently at Resource level. We also need Alarm at Service Level. - ISA-1319 Performance ABE is a pattern, and we need performance at Resource and Service Level.

Open API - AP-4257 TMF649 is in preview state. We need to publish an API to set Thresholds for Performance Management. - TMFC037 and TMFC038 currently expose the same API -

TMF628 Performance Management API - which violates the ODA component principle stating that a single Open API specification must be exposed by only one ODA component. This issue needs to be addressed.

5.3. Further resources

- IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	13-Apr-2023	Amaia White	Final edits prior to publication
1.0.1	25-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	14-Aug-2023	Amaia White	Final edits prior to publication
1.1.0	13-Aug-2024	Milind Bhagwat	Moved to new template
1.1.0	06-Sep-2024	Amaia White	Final edits prior to publication
1.2.0	17-Dec-2024	Milind Bhagwat	Update to remove event API
1.2.0	27-Dec-2024	Rosie Wilson	Final edits prior to publication
1.2.1	03-Jun-2025	Gaetano Biancardi	TMF638 Service Inventory Management API: diagram correction to mark it as mandatory (was already mandatory in the table)
1.2.1	15-Jul-2025	Milind Bhagwat	Added new eTOM business activities; updated table and diagrams for API versions; responded to some comments

1.2.1	15-Sep-2025	Rosie Wilson	Updated Maturity Level to Beta as asset contains Preview APIs; final edits prior to publication
1.3.0	05-Jun-2026	Hugo Vaughan	Gen 5 API updates
1.3.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	13-Apr-2023	Amaia White	First release
Pre-production	15-May-2023	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	14-Aug-2023	Amaia White	New release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	06-Sep-2024	Amaia White	New release 1.1.0
Production	01-Nov-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	18-Dec-2024	Rosie Wilson	New release 1.2.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	15-Sep-2025	Rosie Wilson	New release 1.2.1
Pre-production	25-Jun-2026	Sannah Sibaya	New release 1.3.0

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team.

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